



2. Introduction and Background

Week 2 focused on developing two console-based Python applications: an **Electricity Bill Calculator** and an **ATM Transaction Simulator**. The programs apply core Python concepts such as conditional logic, functions, dictionaries, and menu-driven control flow, using in-memory data storage and a consistent boxed terminal interface.

3. Objectives of the Project

3a. Objectives - Program 1: Electricity Bill Calculator

- Develop a menu-driven electricity bill calculator.
- Implement slab-wise billing logic.
- Apply discount and fixed service charge rules.
- Store user records using dictionaries.
- Validate user input.

3b. Objectives - Program 2: ATM Transaction Simulator

- Develop a console-based ATM simulator.
- Implement user authentication.
- Support balance enquiry, withdrawal, deposit, and PIN change.
- Store account details using dictionaries.
- Test the program with valid and invalid inputs.

4. Problem Statement / Driving Question

4a. Program 1: Electricity Bill Calculator

How can a terminal application accurately calculate electricity bills using slab-wise tariffs, discounts, and fixed charges while managing multiple user records?

4b. Program 2: ATM Transaction Simulator

How can common ATM operations be simulated securely in a console application while handling invalid credentials and transactions?